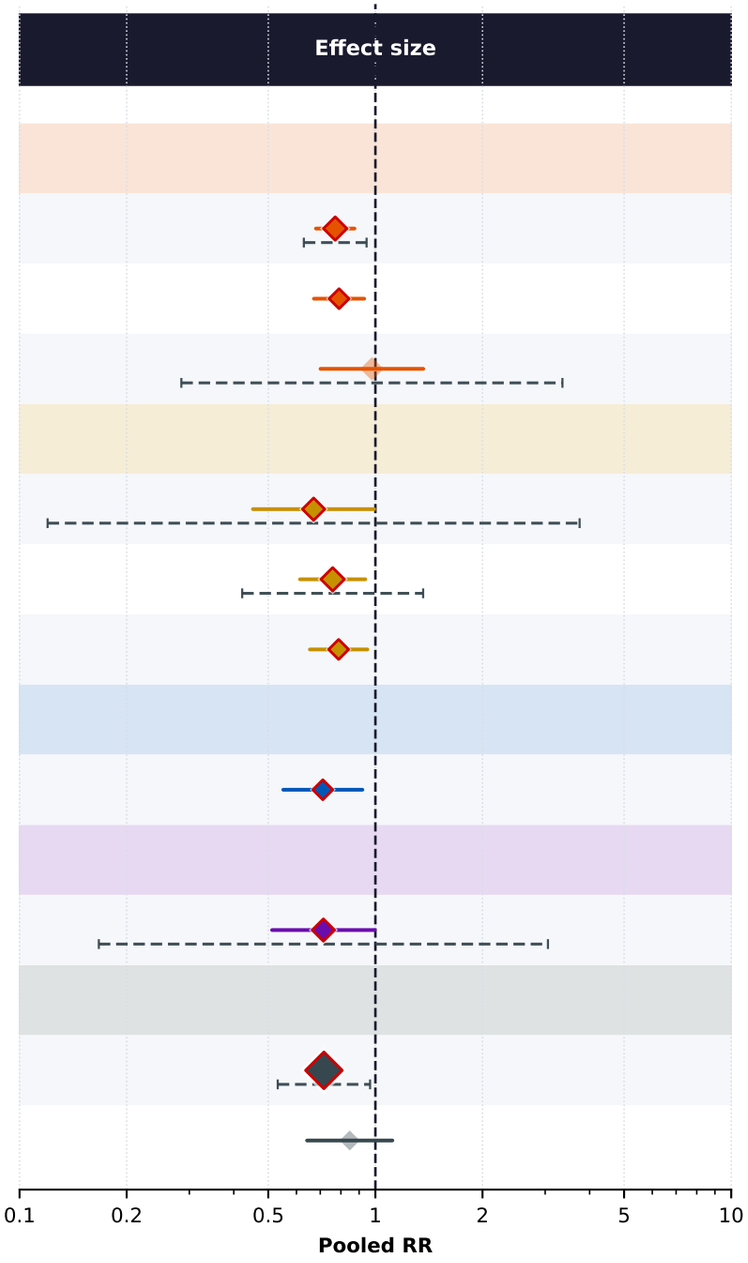
# Exposures negatively associated with renal cell carcinoma risk
## Meta-analysis of dietary-intake studies

| Exposure | Estimates | N | Cases | Pooled RR (95% CI) |
|---|---|---|---|---|
| ***CAROTENOIDS*** | | | | |
| **Lutein** | 5 | 881,922 | 4,765 | *0.77 (0.68-0.87)* |
| **Zeaxanthin** | 2 | 781,129 | 2,616 | *0.79 (0.67-0.93)* |
| **Beta Carotene** | 5 | 881,922 | 4,765 | *0.98 (0.70-1.36)* |
| ***VITAMINS A, C, D, E, K*** | | | | |
| **Vitamin E** | 4 | 127,855 | 2,404 | *0.67 (0.45-0.99)* |
| **Vitamin C** | 5 | 128,222 | 2,559 | *0.76 (0.61-0.94)* |
| **Vitamin D** | 2 | 4,112 | 767 | *0.79 (0.65-0.95)* |
| ***B VITAMINS*** | | | | |
| **Vitamin B1** | 2 | 4,597 | 1,909 | *0.71 (0.55-0.92)* |
| ***POLYPHENOLS & FLAVONOIDS*** | | | | |
| **Tea** | 4 | 804,643 | 2,861 | *0.71 (0.51-1.00)* |
| ***OTHER*** | | | | |
| **Alcohol** | 21 | 3,860,985 | 11,926 | *0.72 (0.65-0.79)* |
| **Total Dietary Fiber Intake** | 2 | 4,410 | 1,621 | *0.85 (0.64-1.12)* |




Exposure group

Carotenoids | Vitamins A, C, D, E, K | B Vitamins | Polyphenols & Flavonoids | Other

*RR < 1 = negatively associated with renal cell carcinoma risk. | ◆ pooled RR (size reflects number of estimates) | ◇ red outline = statistically significant (95% CI excludes 1.0) | ◆ faded = not statistically significant | dashed line below CI = 95% prediction interval (where available) | N and case totals sum estimates and may overlap across strata*

Include subexposures: No